
Linear Statistical Models

B. Statistical Data Science 2nd Year Indian Statistical Institute

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Revision 6 – Positive semi-definite Matrices and Spectral Decomposition

The goal of this set of exercises is to enhance your understanding of positive semi-definite matrices and the important role of spectral decomposition. Recall that a matrix A is called positive semi-definite (p.s.d.) if it is symmetric and $\mathbf{x}^\top A \mathbf{x} \geq 0$ for all $\mathbf{x}$. A matrix A is positive definite (p.d.) if it is p.s.d. and $\mathbf{x}^\top A \mathbf{x} = 0 \Rightarrow \mathbf{x} = 0$.

- (a) Show that if A is p.s.d. (resp., p.d.), then so is cA for any $c > 0$.
- (b) Show that if A and B are p.s.d. (resp., p.d.), then $A + B$ is p.s.d. (resp., p.d.).
- (c) Show that if A is p.s.d. then $CAC^\top$ is p.s.d. for any matrix C . What can we say about positive definiteness of $CAC^\top$?
- (d) Show that for any random vector $\mathbf{Z}$, $\text{Var}(\mathbf{Z})$ is positive semi-definite.

Hint: What is $\mathbf{a}^\top \text{Var}(\mathbf{Z}) \mathbf{a}$ in terms of $\mathbf{Z}$?

Recall that any symmetric matrix A admits a spectral decomposition: $A_{p \times p} = U_{p \times p} \Lambda_{p \times p} U_{p \times p}^\top$, where U is an orthogonal matrix (i.e., $U^\top U = I$) and Λ is a diagonal matrix. Since U is a square matrix, it is non-singular and satisfies $U^{-1} = U^\top$ (why?). The diagonal elements of Λ are the eigenvalues of A and the columns of U are the corresponding (orthogonal) eigenvectors. If A is p.s.d., then all the diagonal elements of Λ (which are the eigenvalues of A) are non-negative. For a p.d. matrix, all the diagonal elements of Λ (which are the eigenvalues of A) are strictly positive.

- (e) Spectral decomposition shows that any p.s.d. matrix can be written as $A = U\Lambda U^\top$, where Λ is a diagonal matrix with non-negative diagonal elements. Show that any matrix of the form $V\Delta V^\top$, where Δ is a diagonal matrix with non-negative diagonal elements, is p.s.d. Hence, a matrix is p.s.d. if and only if it can be written as $U\Lambda U^\top$, where Λ is a diagonal matrix with non-negative diagonal elements. Conclude that a matrix A is p.s.d. if and only if all its eigenvalues are non-negative.
- (f) Repeat the previous exercise for p.d. matrices. Show that a matrix is p.d. if and only if it can be written as $U\Lambda U^\top$, where Λ is a diagonal matrix with positive diagonal elements. Conclude that A is p.d. if and only if all its eigenvalues are positive.
- (g) Show that a matrix A is p.s.d. if and only if it is the variance of some random vector.

Hint: A diagonal matrix with non-negative diagonal elements is the variance of some random vector.

- (h) Let A be an p.s.d. matrix with eigenvalues $\lambda_1, \dots, \lambda_p$. Show that

i. $\text{trace}(A) = \sum_{i=1}^p \lambda_i$.

ii. $\det(A) = \prod_{i=1}^p \lambda_i$.

- (i) Let A be a symmetric matrix with spectral decomposition $A = U\Lambda U^\top$. Show that A is invertible if and only if all the diagonal elements of Λ are non-zero. In this case, $A^{-1} = U\Lambda^{-1}U^\top$.

- (j) Verify that any p.d. matrix is invertible. Moreover, the inverse of a p.d. matrix is also p.d.
- (k) Show that a matrix A is p.d. if and only if it can be written as $A = PP^T$, where P is non-singular. What can we say about p.s.d. matrices?
- (l) For a p.s.d. matrix A with spectral decomposition $U\Lambda U^T$, show that $A^{1/2} = U\Lambda^{1/2}U^T$ satisfies:
- $A^{1/2}$ is p.s.d.
 - $A^{1/2}A^{1/2} = A$.

If A is p.d., then show that $A^{1/2}$ is also p.d.

- (m) Suppose that A has spectral decomposition $A = U\Lambda U^T$. Then, for any λ , $A + \lambda I = U(\Lambda + \lambda I)U^T$ is the spectral decomposition of $A + \lambda I$. So, by adding λI , the eigenvalues of A are increased by λ . Use this to re-verify that for any p.s.d. matrix A and any $\lambda > 0$, $A + \lambda I$ is p.d. What can you say about the form of $(A + \lambda I)^{-1}$? And $(A + \lambda I)^{-1}A(A + \lambda I)^{-1}$?

Hint: Λ is a diagonal matrix.

Recall the Löwner order between matrices. For two symmetric matrices A and B , we say that $A \succeq B$ if $A - B$ is p.s.d. We use $A \succ B$ to denote strict ordering: $A - B$ is p.d.

- (n) Show that if $A \succeq B$ then $CAC^T \succeq CBC^T$ for any matrix C . What happens in the case of strict Löwner order?

Hint: Use (c).

- (o) Show that if $A \succeq B$, then

- $\lambda_{\max}(A) \geq \lambda_{\max}(B)$. ($\lambda_{\max}(A)$ is the largest eigenvalue of A)
- $\lambda_{\min}(A) \geq \lambda_{\min}(B)$. ($\lambda_{\min}(A)$ is the smallest eigenvalue of A)

Hint: $\lambda_{\max}(A) = \max\{\mathbf{x}^T A \mathbf{x} : \|\mathbf{x}\| = 1\}$. $\lambda_{\min}(A) = \max\{\mathbf{x}^T A \mathbf{x} : \|\mathbf{x}\| = 1 \text{ and } \mathbf{x} \perp \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_{d-1}\}\}$, where $\mathbf{v}_i$ is an orthogonal eigenvector of A corresponding to the i -th largest eigenvalue.

- $\text{trace}(A) \geq \text{trace}(B)$.
- $\det(A) \geq \det(B)$.

Hint: Use part (h).

- (p) Let $A \succeq I$. Verify that A is p.d. and show that $A^{-1} \preceq I$.

Hint: Use (n) with (l) and (i).

- (q) Suppose that A and B are p.d. matrices satisfying $A \succeq B$. Then, show that $A^{-1} \preceq B^{-1}$.

Hint: Use the previous part along with (i) and (l).

Spectral decomposition can be stated in a different way as follows: For any symmetric matrix A , there exists an orthogonal matrix U such that $U^T A U = \Lambda$, where Λ is a diagonal matrix. In other words, any symmetric matrix A is “diagonalizable”. We say that two matrices A and B are “simultaneously diagonalizable” if there exists an orthogonal matrix U such that $U^T A U = \Lambda$ and $U^T B U = \Delta$, where both Λ and Δ are diagonal matrices.

- (r) Show that any p.s.d. matrix and I are simultaneously diagonalizable.

Hint: Spectral decomposition.

- (s) Suppose that A and B are two p.s.d. matrices which are simultaneously diagonalizable. Find the condition under which $A \succeq B$.
- (t) Show that for any p.d. matrix A , A^{-1} and $(A + \lambda I)^{-1} A (A + \lambda I)^{-1}$ are simultaneously diagonalizable. Verify that $A^{-1} \succeq (A + \lambda I)^{-1} A (A + \lambda I)^{-1}$.